

Foundation Model

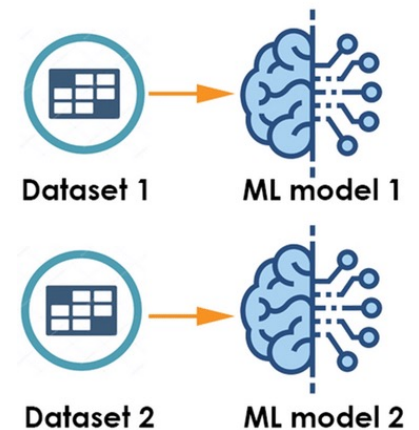
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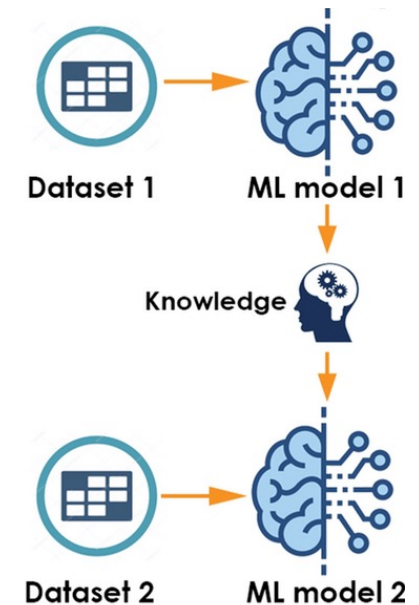
Training from Scratch

- Training from scratch: train a model from initial random weights with no prior knowledge.
 - Typically requires a large amount of labeled data.
 - Usually requires a lot of time and computational resources.
 - Provides flexibility to tailor the model architecture and training process to the specific problem.
 - Often used when little to no relevant pre-trained models are available or when very specific tasks need unique model architectures that existing models cannot accommodate.



Transfer Learning

- Transfer learning: taking a **pre-trained model** that was originally trained on a different, typically larger and more general dataset, and **adapting** it to a specific target task.
- Types:
 - **Feature Extraction**: Use the pre-trained model to extract features and train just a new classifier on top.
 - **Fine-Tuning**: Retrain some or all of the layers of the pre-trained model along with additional layers specific to the task.

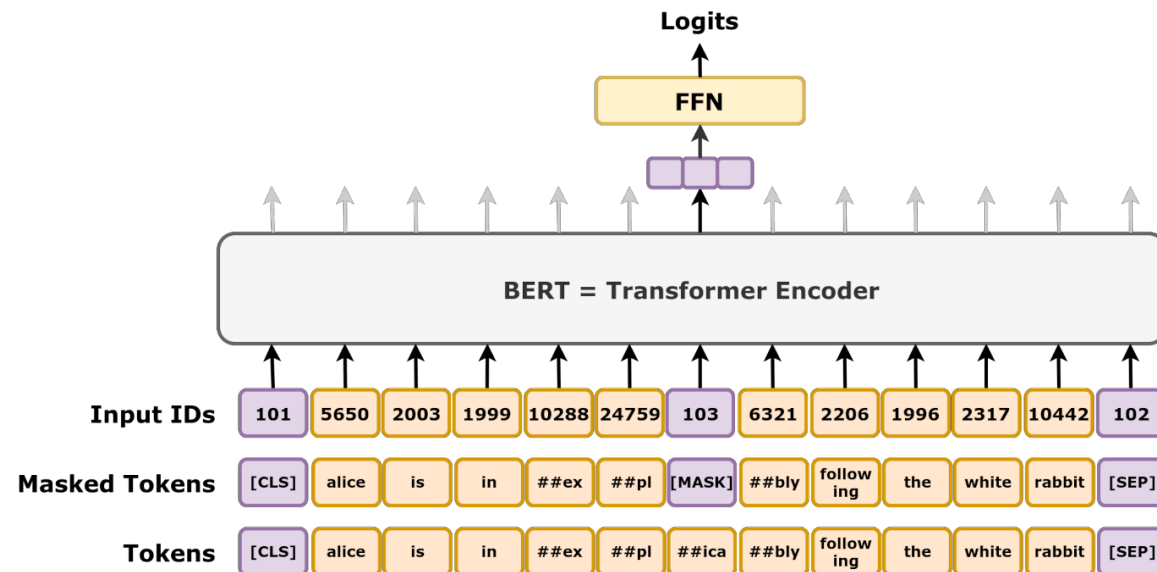


Transfer Learning with Transformers

- Transformer models are often used as a foundation for transfer learning.
 - Transfer learning with Transformers typically involves using pre-trained models that have been trained on large datasets and then fine-tuning them on specific tasks with smaller, task-specific datasets.
- Types:
 - Encoder only: BERT
 - Encoder-decoder: T5
 - Decoder only: GPT

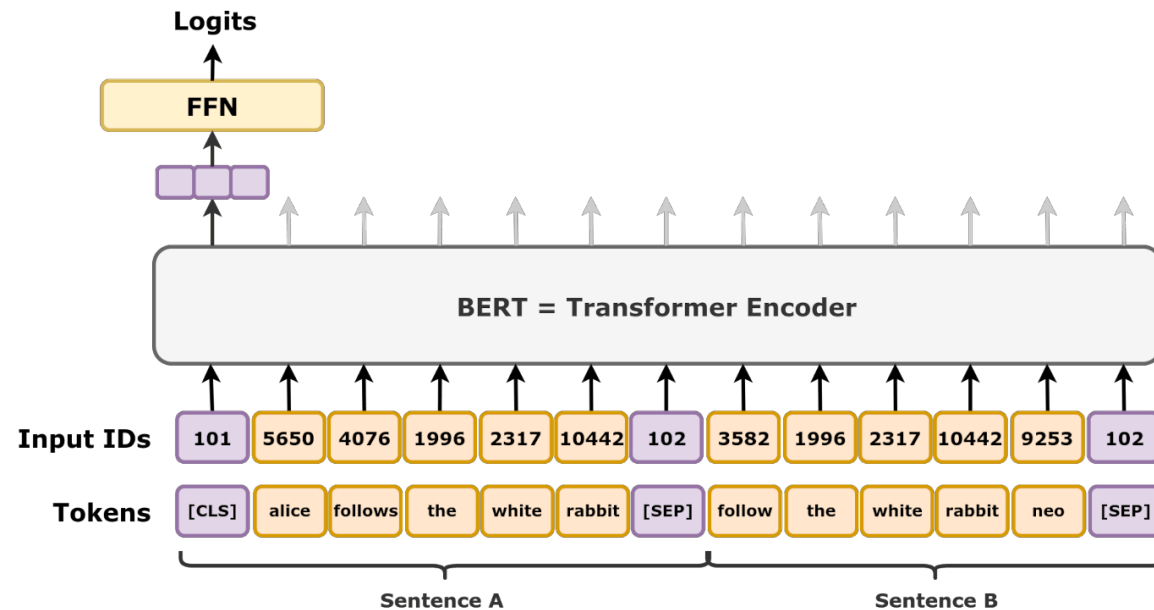
BERT: Pre-Training

- Task 1: masked language modeling
 - Mask out some tokens randomly, and the training objective is to predict them given the context.



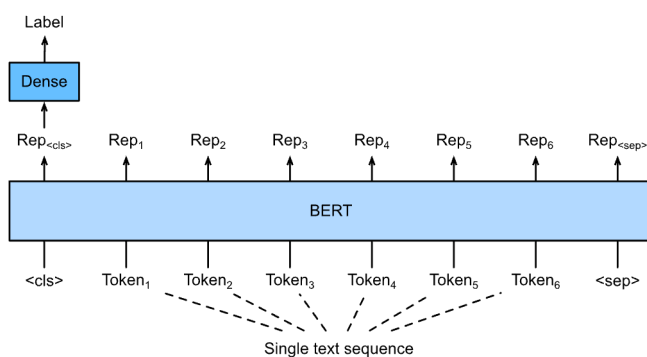
BERT: Pre-Training

- Task 2: next sentence prediction
 - Given two spans of text, the model predicts if these two spans appeared sequentially in the training corpus, outputting either [IsNext] or [NotNext].

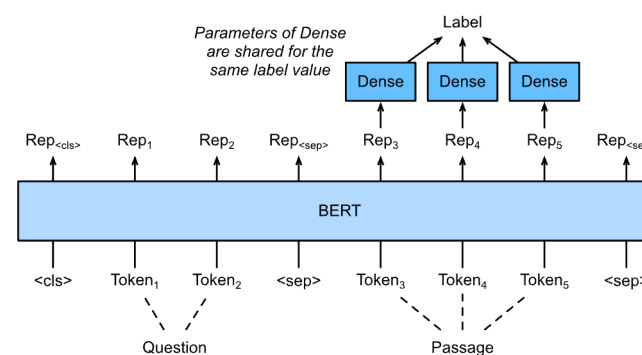


BERT: Fine-Tuning

- After pre-training, BERT can be fine-tuned with fewer resources on smaller datasets to optimize its performance on specific tasks such as natural language inference and text classification, and sequence-to-sequence-based language generation tasks such as question answering and conversational response generation.



Sentence classification

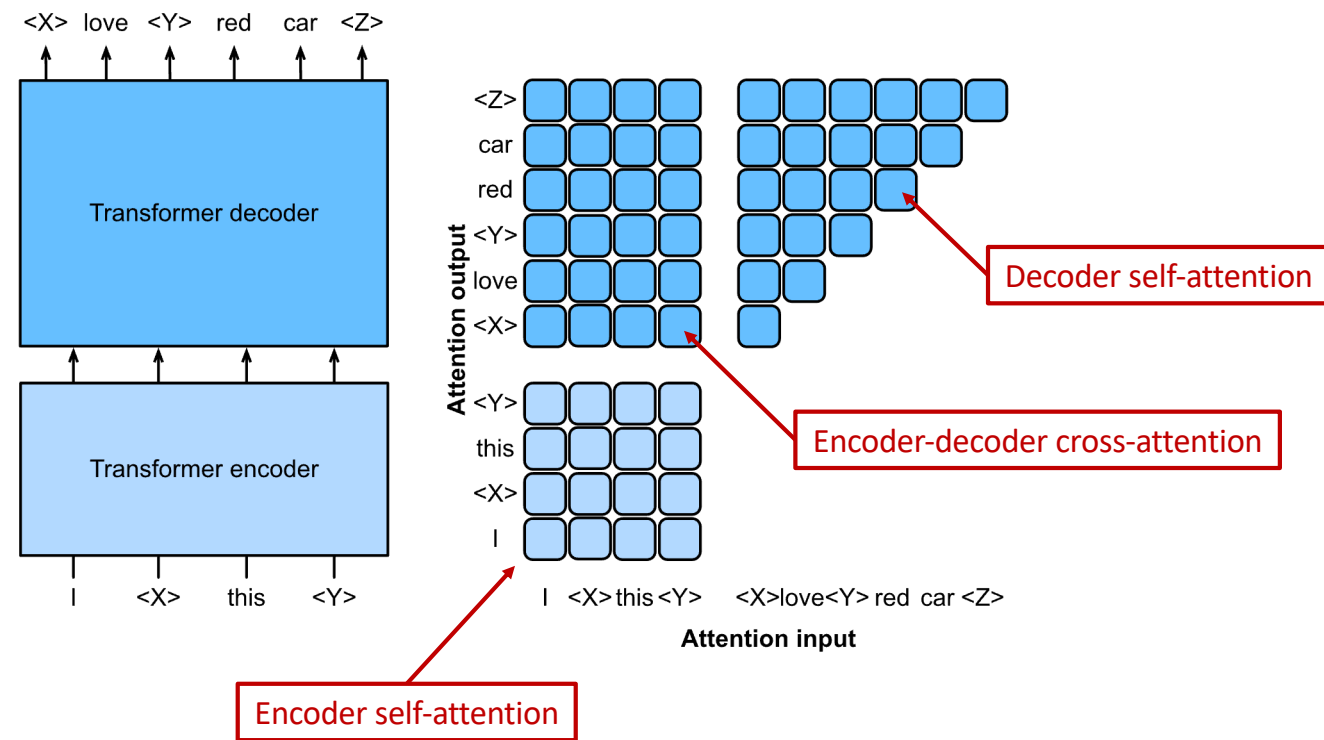


Answering multiple-choice questions

T5

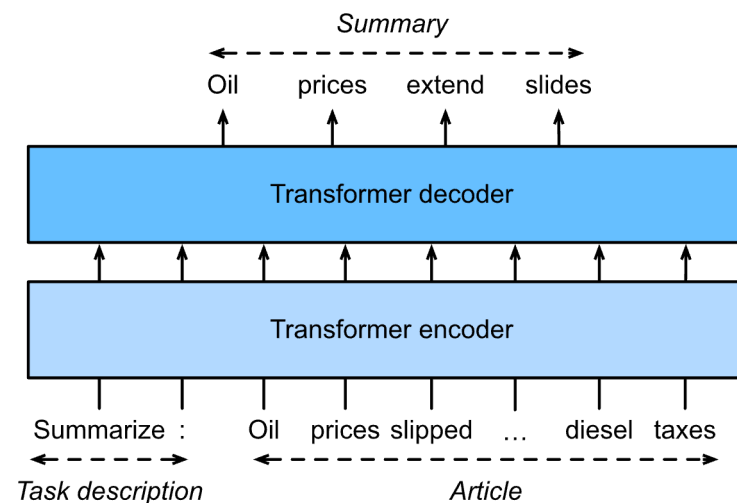
- T5 (Text-to-Text Transfer Transformer)
 - Since a Transformer encoder converts a sequence of input tokens into the same number of output representations, the encoder-only mode cannot generate a sequence of arbitrary length as in machine translation.
 - As originally proposed for machine translation, the Transformer architecture can be outfitted with a decoder that autoregressively predicts the target sequence of arbitrary length, token by token, conditional on both encoder output and decoder output:
 - For conditioning on encoder output, **encoder-decoder cross-attention** allows target tokens to attend to all input tokens;
 - Conditioning on decoder output is achieved by a so-called **causal attention** where any target token can only attend to past and present tokens in the target sequence.

T5: Pre-Training



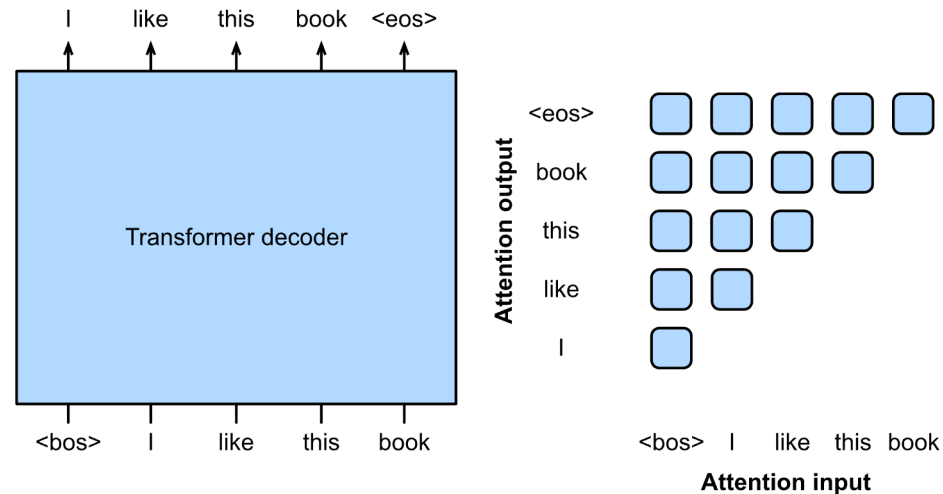
T5: Fine-Tuning

- Major differences of T5 from BERT fine-tuning include:
 - T5 input includes task descriptions;
 - T5 can generate sequences with arbitrary length with its Transformer decoder;
 - No additional layers are required.



GPT

- GPT (Generative Pre-trained Transformer)
 - Decoder-only Transformers remove the entire encoder and the decoder sublayer with the encoder–decoder cross-attention from the original encoder–decoder architecture.

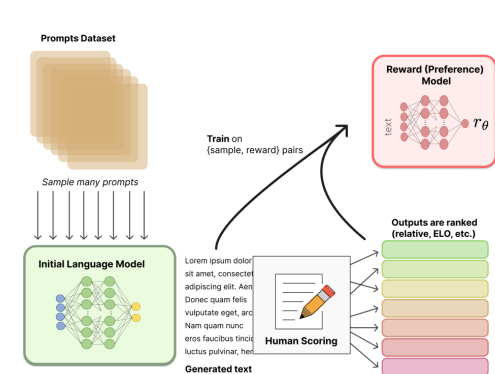


GPT: Training

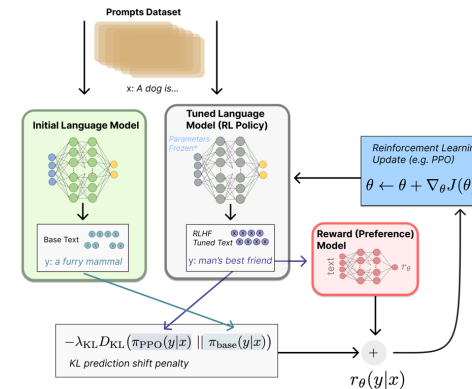
- Language modeling (pre-training):
 - The core task during pre-training is language modeling, where the model learns to predict the next word in a sentence given the preceding words.
 - This is typically done using large-scale datasets collected from the internet, enabling the model to understand grammar, facts about the world, reasoning abilities, and some level of common sense.
- Fine-tuning (post-training):
 - Fine-tuning involves taking the pre-trained model and training it further on a smaller, task-specific dataset to adapt it to particular tasks.
 - This is sometimes supervised learning, where labeled data is used to guide the model's adjustments to become more proficient at specific applications like summarization, translation, sentiment analysis, etc.

GPT: Training

- Reinforcement Learning from Human Feedback (RLHF):
 - Refine the model's outputs to better align with human preferences and ethical considerations.
 - This is often done through the use of a reward model trained on human feedback, guiding the model to produce outputs that humans find helpful or safe.



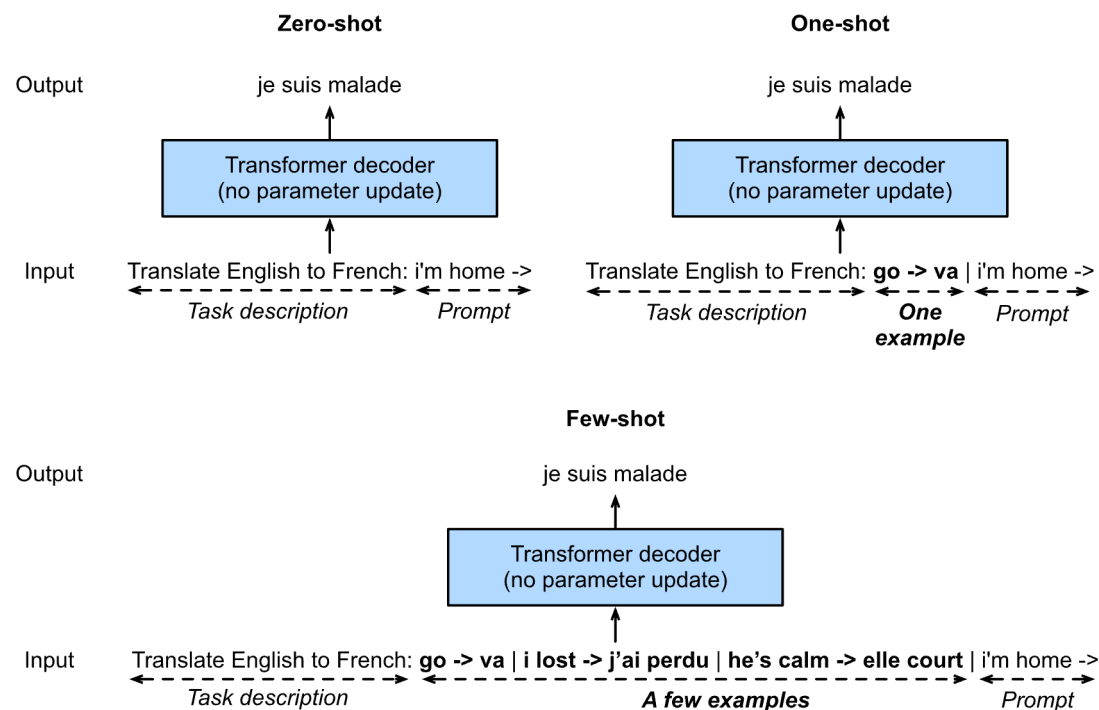
Train a reward model to predict the feedback scores

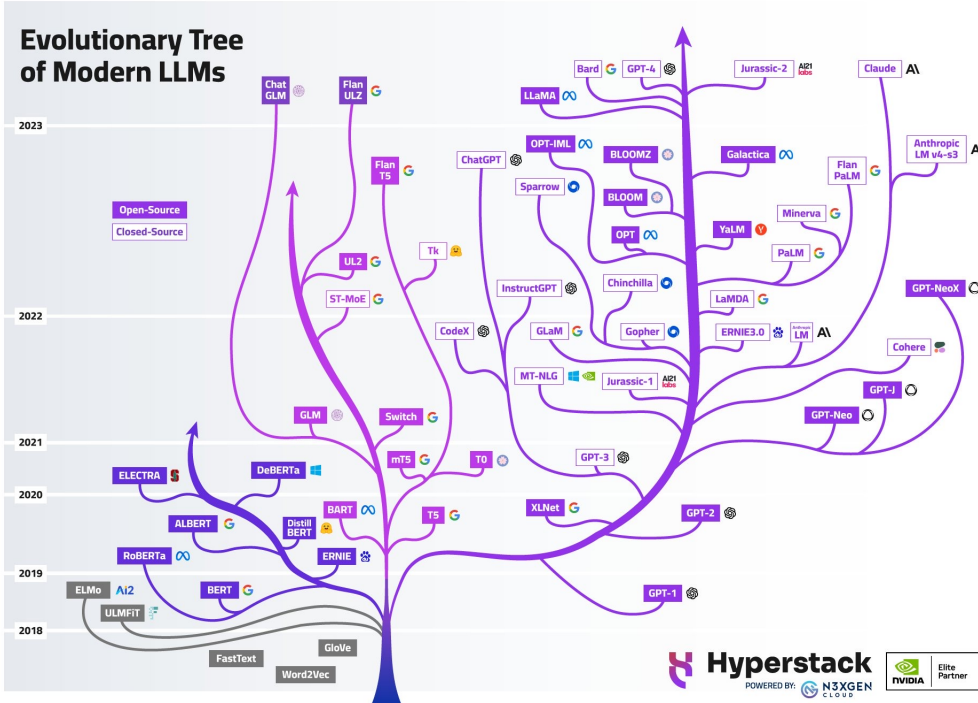


Optimize the language model's policy (how it generates text) based on the rewards

In-Context Learning

- A pretrained language model may generate the task output as a sequence without parameter update, conditional on an input sequence with the task description, task-specific input–output examples, and a prompt (task input).
- In-context learning can be utilized for various tasks such as translation, summarization, question-answering, or even math problem-solving. Users can provide a few examples or a clear instruction to see how the model handles similar cases.





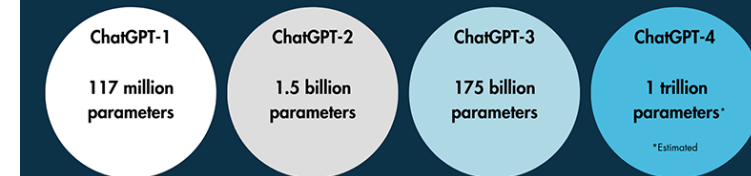
- Word2Vec: introduce the concept of word embedding;
- BERT: breakthrough from Google to understand the context of a word;
- GPT-1: introduce the concept of generative pre-training for language tasks;
- GPT-2: more powerful to generate coherent and contextually relevant text;
- GPT-3: further push the boundaries of what is possible with language models;
- ChatGPT;
- LLaMA;
- Claude;
- GLM;
- ...

<https://medium.com/state-of-the-art-technology/llm-evolution-c17a513f0645>

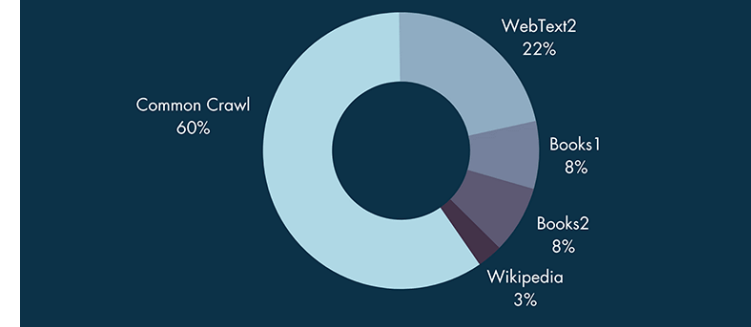
OpenAI's GPT-n series

Model	Architecture	Parameter count	Training data	Release date	Training cost
GPT-1	12-level, 12-headed Transformer decoder (no encoder), followed by linear-softmax.	117 million	BookCorpus: ^[39] 4.5 GB of text, from 7000 unpublished books of various genres.	June 11, 2018 ^[9]	30 days on 8 P600 GPUs, or 1 petaFLOP/s-day. ^[9]
GPT-2	GPT-1, but with modified normalization	1.5 billion	WebText: 40 GB of text, 8 million documents, from 45 million webpages upvoted on Reddit.	February 14, 2019 (initial/limited version) and November 5, 2019 (full version) ^[40]	"tens of petaflop/s-day", ^[41] or 1.5e21 FLOP. ^[42]
GPT-3	GPT-2, but with modification to allow larger scaling	175 billion ^[43]	499 billion tokens consisting of CommonCrawl (570 GB), WebText, English Wikipedia, and two books corpora (Books1 and Books2).	May 28, 2020 ^[41]	3640 petaflop/s-day (Table D.1 ^[41]), or 3.1e23 FLOP. ^[42]
GPT-3.5	Undisclosed	175 billion ^[43]	Undisclosed	March 15, 2022	Undisclosed
GPT-4	Also trained with both text prediction and RLHF; accepts both text and images as input. Further details are not public. ^[38]	Undisclosed. Estimated 1.7 trillion. ^[44]	Undisclosed	March 14, 2023	Undisclosed. Estimated 2.1×10^{25} FLOP. ^[42]

ChatGPT training dataset size



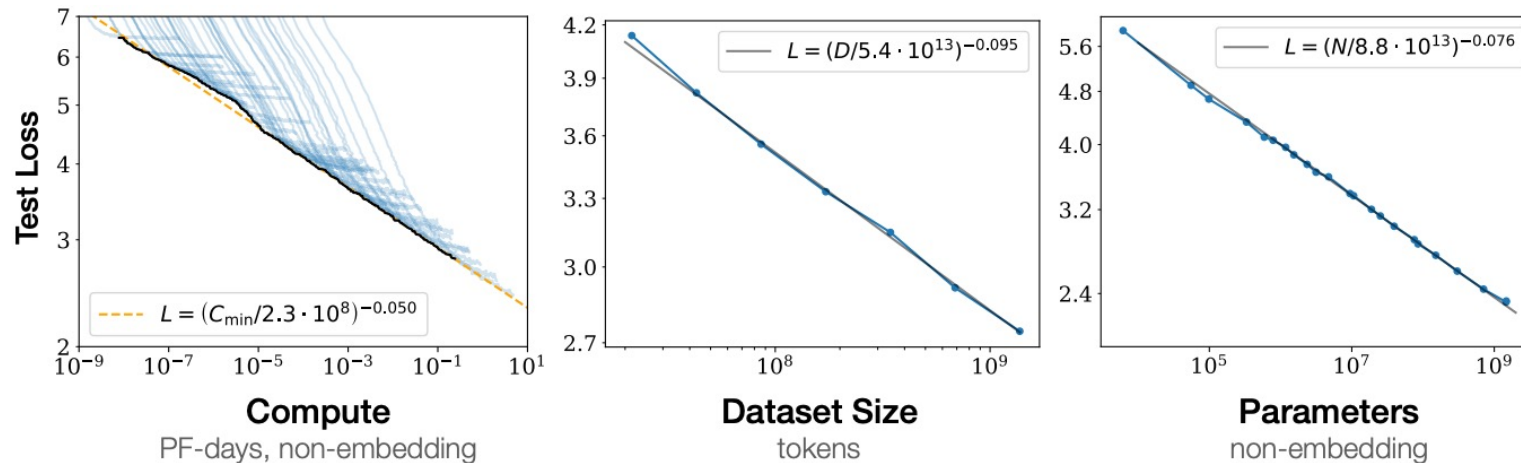
ChatGPT-3 training dataset sources



https://en.wikipedia.org/wiki/Generative_pre-trained_transformer

Scaling Law

- Empirical performance has a power-law relationship with the **amount of training compute** (PetaFLOP/s-days), **dataset size** (number of training tokens) and **model size** (number of parameters).



LLaMA

- LLaMA (Large Language Model Meta AI)

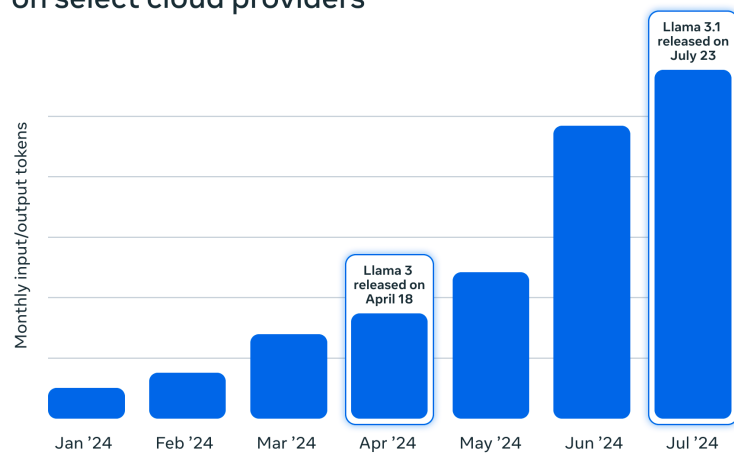
Name ↕	Release date ↕	Parameters ↕	Training cost (petaFLOP-day) ↕	Context length (tokens) ↕	Corpus size (tokens) ↕	Commercial viability? ↕
LLaMA	February 24, 2023	• 6.7B	6,300 ^[35]	2048	1–1.4T	No
		• 13B				
		• 32.5B				
		• 65.2B				
Llama 2	July 18, 2023	• 6.7B	21,000 ^[36]	4096	2T	Yes, subject to acceptable use policy
• 13B						
Code Llama	August 24, 2023	• 69B				
		• 6.7B				
		• 13B				
Llama 3	April 18, 2024	• 33.7B	100,000 ^{[37][38]}	8192	15T	
		• 69B				
Llama 3.1	July 23, 2024	• 8B	440,000 ^{[34][39]}	128,000		
• 70.6B						
Llama 3.2	September 25, 2024	• 8B	128,000 ^[42]			
Llama 3.3	December 7, 2024	• 70.6B				
		• 405B				
		• 1B				
		• 3B				
		• 11B				
		• 90B ^{[40][41]}				
		• 70B		128,000		

Category Benchmark	Llama 3.1 405B	Nemotron 4 340B Instruct	GPT-4 (0125)	GPT-4 Omni	Claude 3.5 Sonnet
General					
MMLU (0-shot, CoT)	88.6	78.7 (non-CoT)	85.4	88.7	88.3
MMLU PRO (5-shot, CoT)	73.3	62.7	64.8	74.0	77.0
IFEval	88.6	85.1	84.3	85.6	88.0
Code					
HumanEval (0-shot)	89.0	73.2	86.6	90.2	92.0
MBPP EvalPlus (base) (0-shot)	88.6	72.8	83.6	87.8	90.5
Math					
GSM8K (8-shot, CoT)	96.8	92.3 (0-shot)	94.2	96.1	96.4 (0-shot)
MATH (0-shot, CoT)	73.8	41.1	64.5	76.6	71.1
Reasoning					
ARC Challenge (0-shot)	96.9	94.6	96.4	96.7	96.7
GPQA (0-shot, CoT)	51.1	-	41.4	53.6	59.4
Tool use					
BFCL	88.5	86.5	88.3	80.5	90.2
Nexus	58.7	-	50.3	56.1	45.7
Long context					
ZeroSCROLLS/QuALITY	95.2	-	95.2	90.5	90.5
InfiniteBench/En.MC	83.4	-	72.1	82.5	-
NIH/Multi-needle	98.1	-	100.0	100.0	90.8
Multilingual					
Multilingual MGSM (0-shot)	91.6	-	85.9	90.5	91.6

LLaMA

- <https://github.com/meta-llama/llama3>
- <https://huggingface.co/meta-llama>

Hosted API usage of Llama
on select cloud providers



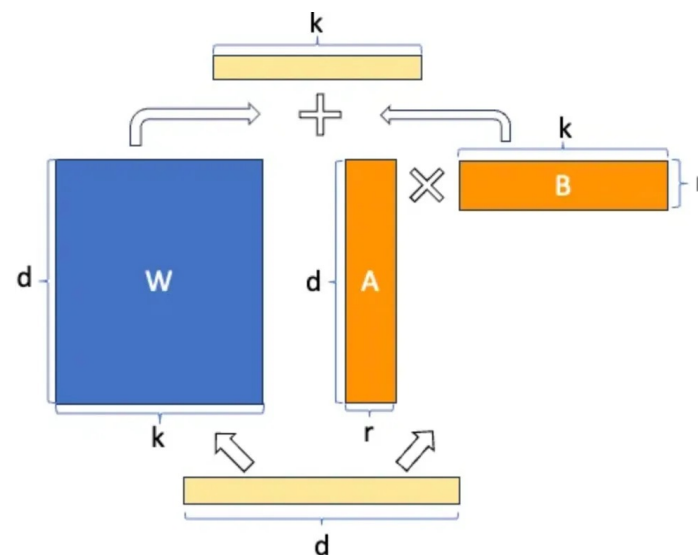
Parametric-Efficient Fine-Tuning (PEFT)

- Full fine-tuning: updates all of the parameters of the pre-trained model.
 - Provides potentially the best performance by fully adapting the model to the new task.
 - Incurs greater risk of overfitting, especially with small task-specific datasets, because the entire model adapts to the new data.
 - Requires significant computational resources, including high memory requirements and longer training times, as all parameters are being adjusted.
- PEFT: modifies only a small subset of the model's parameters.
 - Requires less powerful hardware and can be quicker to train.
 - Offers a good trade-off between performance and resource use.

PEFT: LoRA

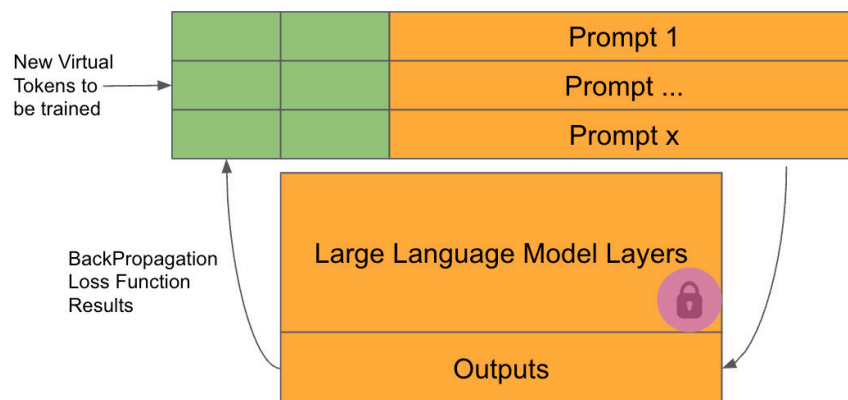
- Low-rank adaptation:
 - This technique involves approximating the weight matrix updates with low-rank matrices.

1. Freezing the original model weights;
2. Calculate changes in a separate set of weights;
3. Add to the original parameters.



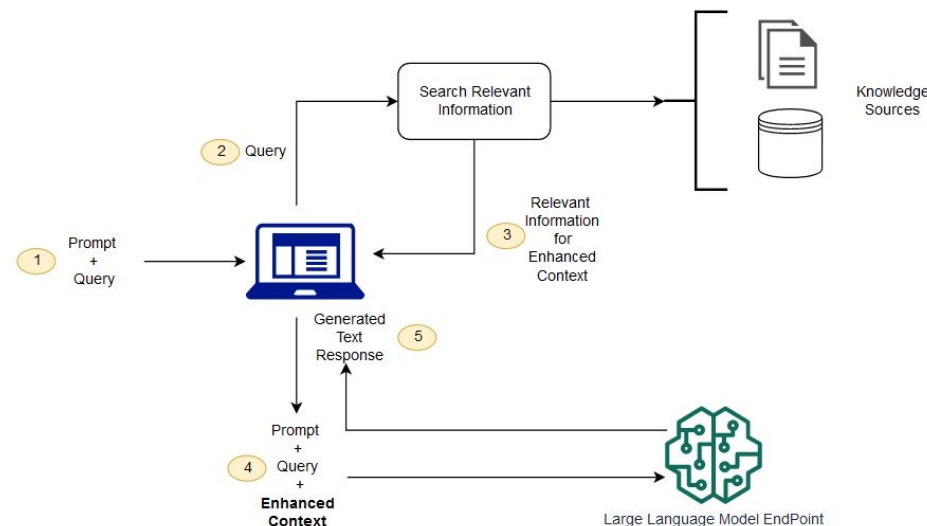
PEFT: Prompt Tuning

- Prompt tuning:
 - Instead of adjusting model weights, this approach involves prepending task-specific tokens or embeddings to the input.
 - This allows models to adapt to new tasks by altering input representations.

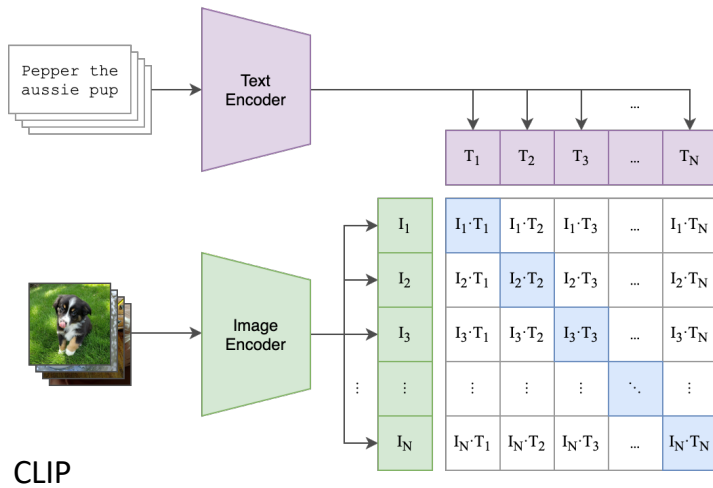
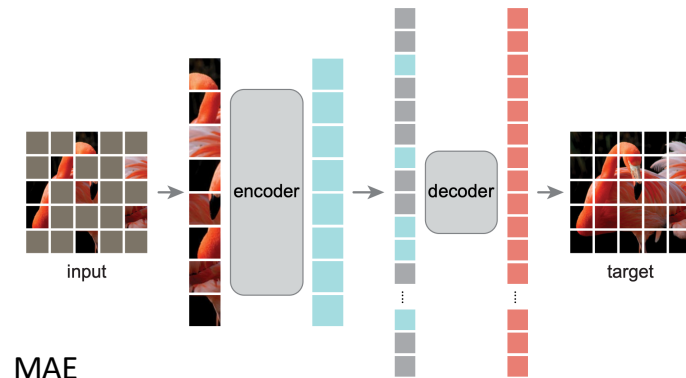


Retrieval-Augmented Generation (RAG)

- RAG is an advanced technique designed to enhance the capabilities of language models by combining the strengths of retrieval-based and generative approaches.



Visual Foundation Model



Model	Provider	Task	Inputs	Training Cost	Inference Speed (1 NVIDIA T4, PyTorch)	License
CLIP	OpenAI	Embedding Extraction	Images and Text	None	Varies depending on encoder and modality. Efficient implementations can achieve sub-5 ms/image latency	Open Source
DINOv2	Meta AI	Embedding Extraction	Images	None	Same as above	Non-Commercial
ImageBind	Meta AI	Embedding Extraction	Images, Text, Audio, Depth Maps, Thermal Image Maps, Inertial Measurement Units	None	Same as above	Non-Commercial
SAM	Meta AI	Instance Segmentation	Images and Point Prompts	None	2 images/s	Open Source
OWL-ViT	Google	Open-Set Object Detection	Images and Text Queries	None	100 images/s	Open Source
YOLO-NAS	Deci.ai	Object Detection	Images	Moderate computational cost. Moderate data cost. Requires labelled data	4 images/s	Code is Open Source, Pre-trained weights are Non-Commercial.
Mask2Former	Meta AI	Panoptic, Instance, and Semantic Segmentation	Images	Moderate computational cost. High data cost. Requires labelled data	2 s/image	Open Source
DETR	High	Object Detection	Images	Moderate computational cost. Low data cost. Requires labelled data	40 ms/image	Open Source
ConvNeXt	Meta AI	Classification, Object Detection, Image Segmentation	Images	High computational cost. Moderate data cost. Requires labelled data	10 ms/image	Open Source

Multi-Modal Integration



What recipe can you make with these ingredients?



Based on the ingredients visible in the image, you can make a classic pasta dish with tomato sauce. Here's a simple recipe you could follow:

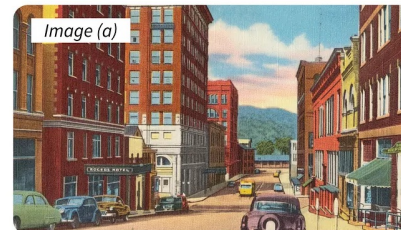
Ingredients:

- Pasta (such as spaghetti or any other type you prefer)
- Fresh tomatoes
- [...] Garlic

Instructions:

1. **Prepare the Ingredients:**
 - Dice the onion.
 - [...]
2. **Cook the Pasta:**
 - Bring a large pot of salted water to a boil.
 - [...]
3. **Make the Tomato Sauce:**
 - In a large pan, heat some olive oil over medium heat.
 - [...]

You are an expert in art. Provide a descriptive caption for the painting.



A colorful depiction of a mid-20th century main street at dusk, with classic cars parked alongside period architecture under a pastel sunset sky

Thank You!